

Homework 1

Due Date: End of Week 2: Sunday @ Midnight

Email to: twalmsle@usc.edu

Question 1

Assume an economy with three markets with supply and demand curves for three goods given by:

$$q_1^s = -20 + p_1$$

$$q_2^s = -100 + 2p_2$$

$$q_3^s = p_3$$

$$q_1^d = 80 - 2p_1 - p_3$$

$$q_2^d = 200 - p_2$$

$$q_3^d = 100 - 2p_3 - p_1$$

- Comment on the relationships between demand and supply of the three goods.
- What are the equilibrium conditions for the three goods?
- Solve for the equilibrium prices and quantities of the three goods.

Question 2

An input-output (IO) table depicts the transactions made in a country/economy, and a multi-regional input-output (MRIO) table depicts the transactions within and between multiple countries. In the simple MRIO table below we have two countries (D and F) and two goods. Country D produces both goods X_{1D} and X_{2D} and country F produces only one good X_{1F} . The commodities are labelled X_1 to X_3 to keep things simpler.

Down the columns of an IO (or MRIO) table we see how goods are produced, i.e., by combining intermediate inputs (yellow) and value added (orange). For example, the good, orange juice, is produced by combining oranges and water in plastic bottles from abroad (intermediate inputs, yellow) using labor and a juicing machine (value added, orange). The rows show the sales of each commodity to other industries for the production of other goods (yellow)(e.g., OJ might be used in production of restaurant meals) and sales to final consumers in countries D (c_D) and F (c_F) (green). Also note that total sales (row totals, blue) for a product must equal total production (column totals, blue) (i.e., demand must equal supply).

MRIO table for Countries D and F

			Goods			Final consumption		Total sales
			X _{1D}	X _{2D}	X _{1F}	C _D	C _F	
			X ₁	X ₂	X ₃			
Goods	X _{1D}	X ₁	40	10	0	110	40	200
	X _{2D}	X ₂	0	30	0	60	10	100

	X _{1F}	X ₃	10	0	30	40	70	150
Payments to labor and owners	Value added		150	60	120			
Total production			200	100	150			

Leontief recognized that output of each commodity X_i , where $i = 1...3$, should equal intermediate plus final demand:

$$X_i = \sum_{j=1}^3 N_{i,j} + \sum_{c=1}^2 C_{i,c} \quad (1)$$

Where: $N_{i,j}$ is (intermediate) demand for commodity i in production of commodity j . e.g. Good 1 (first row) used in production of good 2 (second column), $N_{1,2} = 10$

$C_{i,c}$ is final demand for good i , by consumers in country c . e.g., final demand for good 1 by Country F, $C_{1,F} = 40$.

For example, for Good 1, total production equals the sum of any good 1 used in production of goods 1, 2 and 3 plus final demand by Countries D and F:

$$\begin{aligned} X_1 &= N_{1,1} + N_{1,2} + N_{1,3} + C_{1,D} + C_{1,F} \\ 200 &= 40 + 10 + 0 + 110 + 40 \end{aligned}$$

Leontief also recognized that intermediate inputs depended on the amount to be produced. Assuming intermediate inputs are in fixed proportions, then demand for intermediate input i by producers of good j would always be a fixed proportion $a_{i,j}$ of production of commodity j (X_j), hence:

$$N_{i,j} = a_{i,j} \cdot X_j \quad (2)$$

Where: the parameters $a_{i,j}$ are derived from the MRIO table. For example: $a_{1,2} = \frac{N_{1,2}}{X_2} = \frac{10}{100} = 0.1$,

where 100 is the total sum for column or row X_2 (totals of rows and columns should equal in an MRIO)

Put another way, if 10 units of good 1 are needed to produce 100 units of good 2, then 30 units of good 1 would be needed to produce 300 units of good 2 – the ratio of inputs to output is always 0.1.

Hence:

$$X_i = \sum_{j=1}^3 a_{i,j} \cdot X_j + \sum_{c=1}^2 C_{i,c} \quad (3)$$

Example for good 1:

$$\begin{aligned} X_1 &= a_{1,1}X_1 + a_{1,2}X_2 + a_{1,3}X_3 + a_{1,4}X_4 + C_{1,D} + C_{1,F} \\ X_1 &= 0.2X_1 + 0.1X_2 + 0X_3 + 0X_4 + C_{1,D} + C_{1,F} \\ 200 &= 0.2 \times 200 + 0.1 \times 100 + 0 \times 150 + 0 \times 60 + 110 + 40 \end{aligned}$$

This shows that production of good i depends not only on final demand for good i ($C_{i,c}$), but also on the demand for good i as an intermediate in the production of all other goods j (X_j). Hence, while final

demand for good 1 is 150 units (110+40), 200 units of good 1 must be produced in order to produce goods 1 and 2 (and 3).

The matrix of $a_{i,j}$'s is usually labelled A and is called the technology matrix, as it shows the inputs (technology) needed to produce each good.

If you want to read more on this, you can search for Leontief matrix on google, where you will find several youtube videos and wiki pages

Answer the following questions.

a: Write equation (3) in matrix algebra using $\mathbf{x}$ for the vector of production, A as the matrix of parameters $a_{i,j}$ and vectors $\mathbf{c}_D$ and $\mathbf{c}_F$ representing the final demands of country D and F.

b: Calculate the entries of the A matrix, also known as the technology matrix.

c: Starting from the equation derived in part a, derive $\mathbf{x}$ as a function of A, $\mathbf{c}_D$ and $\mathbf{c}_F$ in matrix algebra notation (i.e. $\mathbf{x}$ should appear only on the left-hand side of the equation).

d: In your answer to question part c, you should see the following term within the equation: $[\mathbf{I} - \mathbf{A}]^{-1}$. This matrix is referred to as the Leontief matrix $\mathbf{L} = [\mathbf{I} - \mathbf{A}]^{-1}$. Calculate the Leontief matrix (show working).

e: Using linear algebra show what happens to $\mathbf{x}$ if domestic final consumer demand ($\mathbf{c}_D$) for X_1 increases by 30? Explain why production of X_3 rises?

f: Use the equation from question part c, and the Leontief matrix to determine the amounts produced of each good to meet country D's final demand $\mathbf{c}_D$ (only) (i.e., drop $\mathbf{c}_F$ from the equation)?

Use the initial data when answering this question (i.e., ignore any changes made to demand in part e).

g: If the consumption of X_3 creates emissions of 1 unit of CO_2 for every dollar produced and consumption of X_1 creates 0.5 units of CO_2 (X_2 creates zero emissions). Using linear algebra answer the following:

- i. Create a row vector of emissions for the three commodities? Label the row vector $\mathbf{e}$.
- ii. Calculate the total amount of CO_2 emitted because of the production of all goods? (i.e. calculate $\mathbf{e} \cdot \mathbf{x}$)
- iii. Calculate the amount of CO_2 emitted because the production of good X_1 , X_2 and X_3 ? What is the formula in linear algebra to calculate this? How and why does it differ from the equation $\mathbf{e} \cdot \mathbf{x}$ used in the previous part of this question?
- iv. Using your answer from part f, calculate the amount of CO_2 emitted because of Country D's demand for final consumers?
- v. What do your results suggest about Country D's emissions? Are they a net importer or exporter of emissions?

Use the initial data when answering this question (i.e., ignore any changes made to demand in part e).